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PROGRAM OF INSTRUCTION

POI-MSS

USAREUR-AF

PROGRAM OF INSTRUCTION

Maven Smart System (MSS) Training Program

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

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PROGRAM OF INSTRUCTION

MAVEN SMART SYSTEM (MSS) TRAINING PROGRAM

USAREUR-AF Operational Data Team — C2DAO

Program Designation	MSS-POI-001
Proponent	USAREUR-AF C2DAO
Effective Date	March 2026
Review Cycle	Annual or upon major platform update
Authority	USAREUR-AF G6/G9 Data Governance Directive
Classification	UNCLASSIFIED

CHAPTER 1 — AUTHORITY, PURPOSE, AND SCOPE

1-1. Authority

Published under authority of the USAREUR-AF C2DAO. Establishes the official curriculum, evaluation standards, and resource requirements for all MSS training within the USAREUR-AF AOR.

1-2. Purpose

Defines course content, instructional methods, evaluation criteria, and resource requirements for each training level. Serves as the authoritative reference for course scheduling, instructor preparation, student enrollment, and program assessment.

1-3. Scope

This POI covers eighteen courses in five tiers:

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Tier	Courses	Notes
1 — Foundation	TM-10	Required for all USAREUR-AF personnel
2 — Builder	TM-20	All staff building or maintaining data products
3 — Advanced Builder	TM-30	Data-adjacent specialists and unit data leads
4a — WFF Functional	TM-40A through TM-40F	Warfighting function staff; prereq TM-30
4b — Specialist	TM-40G through TM-40L	Role-specific specialist training; prereq TM-30

1-4. Prerequisite Chain

```
TM-10 (all personnel)
  └── TM-20 (builders)
        └── TM-30 (advanced builders / data-adjacent / WFF functional staff)
              ├── TM-40A through TM-40F (WFF functional tracks —
INT/FIRES/M2/SUST/PROT/MC staff)
              ├── TM-40G (ORSA)
              ├── TM-40H (AI Engineer)
              ├── TM-40I (ML Engineer)
              ├── TM-40J (Program Manager)
              ├── TM-40K (Knowledge Manager)
              └── TM-40L (Software Engineer)
```

1-5. Training Environment

All MSS training is conducted in the **MSS Training Environment** — a dedicated Foundry instance with synthetic operational data. No training is conducted in the production MSS environment. Trainees do not ingest, modify, or export production operational data during training.

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CHAPTER 2 — PROGRAM OVERVIEW

2-1. Course Summary

Course	Title	Tier	Duration	Hours	Prerequisite
TM-10	Maven User	Foundation	1 day	8	None
TM-20	Builder	Builder	5 days	40	TM-10
TM-30	Advanced Builder	Advanced	5 days	40	TM-10, TM-20
TM-40A	Intelligence WFF	WFF Functional	3 days	24	TM-10, TM-20, TM-30 (Required)
TM-40B	Fires WFF	WFF Functional	3 days	24	TM-10, TM-20, TM-30 (Required)
TM-40C	Movement & Maneuver WFF	WFF Functional	3 days	24	TM-10, TM-20, TM-30 (Required)
TM-40D	Sustainment WFF	WFF Functional	3 days	24	TM-10, TM-20, TM-30 (Required)
TM-40E	Protection WFF	WFF Functional	3 days	24	TM-10, TM-20, TM-30 (Required)
TM-40F	Mission Command WFF	WFF Functional	3 days	24	TM-10, TM-20, TM-30 (Required)
TM-40G	ORSA Specialist	Specialist	5 days	40	TM-10, TM-20, TM-30 (Required)
TM-40H	AI Engineer	Specialist	5 days	40	TM-10, TM-20, TM-30 (Required)
TM-40I	ML Engineer	Specialist	5 days	40	TM-10, TM-20, TM-30 (Required)
TM-40J	Program Manager	Specialist	3 days	24	TM-10, TM-20, TM-30 (Required)
TM-40K	Knowledge Manager	Specialist	3 days	24	TM-10, TM-20, TM-30 (Required)
TM-40L	Software Engineer	Specialist	5 days	40	TM-10, TM-20, TM-30 (Required)

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Course	Title	Tier	Duration	Hours	Prerequisite
TM-50G	Advanced ORSA	Advanced Specialist	5 days	40	TM-40G (Required)
TM-50H	Advanced AI Engineer	Advanced Specialist	5 days	40	TM-40H (Required)
TM-50I	Advanced ML Engineer	Advanced Specialist	5 days	40	TM-40I (Required)
TM-50J	Advanced Program Manager	Advanced Specialist	3 days	24	TM-40J (Required)
TM-50K	Advanced Knowledge Manager	Advanced Specialist	3 days	24	TM-40K (Required)
TM-50L	Advanced Software Engineer	Advanced Specialist	5 days	40	TM-40L (Required)

NOTE

TM-40A–F (WFF functional tracks) require TM-10, TM-20, and TM-30 as hard prerequisites. TM-40G–L (specialist tracks) and TM-50G–L (advanced specialist tracks) also require TM-30 as a hard prerequisite. There are NO TM-50A–F tracks.

Total program hours (full progression to any single WFF or specialist track): 133 hours (TM-10 through TM-40G/H/I/L); 97 hours (TM-10 through TM-40A–F) **Total program hours (full progression to any single advanced specialist track):** 173 hours (TM-10 through TM-50G/H/I/L)

2-2. Training Philosophy

MSS training is competency-based, not time-based. Go/No-Go evaluation at the end of each course is the authoritative measure of completion. Training emphasizes:

Principle	Description
Build, break, recover	Lab time includes structured error-recovery before evaluation
Design before tool	TM-30+: trainees document design before opening the tool
Operational context	All scenarios use operationally plausible data and mission contexts
Governed practice	Data governance, naming, and access control are integrated into every lab

2-3. Instructional Methods

Code	Method	Description
LEC	Lecture	Instructor-delivered; students in listening/note-taking role
LAB	Laboratory	Hands-on tool exercise at workstation
DIS	Discussion	Structured facilitated discussion; student contributions required
SEM	Seminar	Small-group intensive; used for safety blocks and design critique
BRF	Brief	Short instructor-led overview or scenario introduction
REV	Review	Structured review of prior material; student Q&A focused
EVAL	Evaluation	Practical exercise; graded; no instructor assistance permitted
WKS	Workshop	Design workshop; student product reviewed by instructor/peers

CHAPTER 3 — BLOCKS OF INSTRUCTION

3-1. TM-10: Maven User

Course length: 1 day (8 hours) | **Evaluation:** Practical exercise (6 tasks, Go/No-Go)

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Block	Title	Hours	Method	Reference
1	MSS Overview and Data Literacy Fundamentals	1.0	LEC	TM-10 Ch 1; Data Literacy Technical Reference Ch 1
2	Login and Navigation: CAC Authentication, Project Access	1.0	LAB	TM-10 Ch 2
3	Workshop Applications: Tables, Filters, Dashboards	1.0	LAB	TM-10 Ch 3
4	Actions: Executing Status Updates and Form Submissions	1.0	LAB	TM-10 Ch 4
5	Contour: Building a Basic Chart and Applying a Filter	1.0	LAB	TM-10 Ch 5
6	Quiver: Exploring Object Types, Filters, and Exporting Views	1.0	LAB	TM-10 Ch 6
7	AIP Interface: Submitting a Query; Understanding AI Output Limitations	0.5	LAB	TM-10 Ch 7
8	Classification Markings and Authorized Export Procedures	0.5	LEC	TM-10 Ch 8
9	Practical Exercise (Evaluated)	1.0	EVAL	TM-10 Practical Exercise Guide
	Total	**8.0**		

3-2. TM-20: Builder

Course length: 5 days (40 hours) | **Evaluation:** Practical exercise (11 tasks, Go/No-Go)

Day 1 — Project Fundamentals and File Ingestion

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Block	Title	Hours	Method	Reference
1	Project Creation: Naming Conventions, Markings, Folder Structure	1.5	LAB	TM-20 Ch 2
2	File Ingestion: Upload CSV, Inspect Schema, Types, Row Count	0.75	LAB	TM-20 Ch 3 Sec 3-1
3	Dataset Explorer: Column Profiling, Null Detection, Type Mismatches	1.0	LAB	TM-20 Ch 3 Sec 3-2
4	Pipeline Builder Orientation: Canvas, Step Library, I/O Config	2.0	LAB	TM-20 Ch 3 Sec 3-3
5	C2DAO Naming Conventions: Datasets, Pipelines, Object Types	0.5	DIS	Standards Ch 1-2
6	Individual Practice: Second Project, Ingest Provided Dataset	1.5	LAB	TM-20 Ch 2-3

Day 2 — Pipeline Builder: Clean and Transform

Block	Title	Hours	Method	Reference
7	Pipeline: Filter Step, Rename Step, CAST for Type Correction	2.0	LAB	TM-20 Ch 3 Sec 3-4
8	Pipeline: Calculated Columns — String Functions, Conditional Logic, COALESCE	1.25	LAB	TM-20 Ch 3 Sec 3-5
9	Pipeline: Date and Time Functions — DATEDIFF, DATE_TRUNC, CURRENT_DATE	2.0	LAB	TM-20 Ch 3 Sec 3-6
10	End-to-End Pipeline Practice: Raw Input to Typed Filtered Output	1.75	LAB	TM-20 Ch 3

Day 3 — Pipeline Builder: Joins; Ontology Manager

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Block	Title	Hours	Method	Reference
11	Pipeline: Join Step — Inner/Left Join, Key Selection, Deduplication	2.0	LAB	TM-20 Ch 3 Sec 3-7
12	Pipeline: Group-By Aggregation, Union Step, Output Mode Configuration	1.25	LAB	TM-20 Ch 3 Sec 3-8
13	Ontology Manager: Create Object Type — Properties, PK, Display Name	2.0	LAB	TM-20 Ch 4 Sec 4-1
14	Ontology Manager: Create Link Type — Cardinality, Directionality	0.75	LAB	TM-20 Ch 4 Sec 4-2
15	Pipeline: Ontology Write Step — Property Mapping, Run and Verify	1.0	LAB	TM-20 Ch 4 Sec 4-3

Day 4 — Actions and Workshop Applications

Block	Title	Hours	Method	Reference
16	Actions: Create Basic Action — Parameter, Write Rule, Access Restriction	1.5	LAB	TM-20 Ch 4 Sec 4-4
17	Workshop Orientation: Canvas, Widget Library, Object Type Binding, Table Widget	1.75	LAB	TM-20 Ch 5 Sec 5-1
18	Workshop: Filter Widget, Metric Widget, Bar Chart Widget	2.0	LAB	TM-20 Ch 5 Sec 5-2
19	Workshop: Connecting Action Button — Trigger, Confirmation, Post-Action Refresh	1.25	LAB	TM-20 Ch 5 Sec 5-3
20	Access Control Model: Viewer vs. Editor Roles	0.5	DIS	TM-20 Ch 6 Sec 6-1

Day 5 — Publishing, Governance, and Practical Exercise

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Block	Title	Hours	Method	Reference
21	Workshop Publishing: Visibility, Viewer Access, Confirm Viewer Cannot Edit	1.0	LAB	TM-20 Ch 6 Sec 6-2
22	Branching: Create Branch, Make Change on Branch, Verify Branch-Only	1.0	LAB	TM-20 Ch 7 Sec 7-1
23	Promotion Workflow: Write Description, Submit to Steward, Respond to Rejection	0.75	LAB	TM-20 Ch 7 Sec 7-2
24	Full-Stack Review: Trace Product from Raw File to Access Control	1.0	REV	TM-20 All Chapters
25	Practical Exercise (Evaluated)	4.0	EVAL	TM-20 Practical Exercise Guide
	TM-20 Total	**40.0**		

3-3. TM-30: Advanced Builder

Course length: 5 days (40 hours) | **Evaluation:** Practical exercise (6 tasks, including reviewed design document; Go/No-Go)

Day 1 — Advanced Workshop

Block	Title	Hours	Method	Reference
1	Multi-Page Workshop: Navigation, Page Parameters, URL Deep Links	2.0	LAB	TM-30 Ch 2 Sec 2-1
2	Conditional Logic: Show/Hide Panels, Conditional Formatting, Dynamic Visibility	1.25	LAB	TM-30 Ch 2 Sec 2-2
3	Variable Passing: Object Selections Between Pages, Filtered Detail Views	2.0	LAB	TM-30 Ch 2 Sec 2-3
4	Design Exercise: 3-Page Operations Dashboard (Portfolio→Unit→Trend); Instructor Critique	1.75	WKS	TM-30 Ch 2

Day 2 — Advanced Pipeline Builder

Block	Title	Hours	Method	Reference
5	Multi-Source Joins: Inner/Left/Outer, Fan-Out Handling, Post-Join Deduplication	2.0	LAB	TM-30 Ch 3 Sec 3-1
6	Union Transforms: Compatible Schemas, Handling Mismatches	1.25	LAB	TM-30 Ch 3 Sec 3-2
7	Group-By Aggregations: Count/Sum/Min/Max, Aggregate-Then-Join Patterns	2.0	LAB	TM-30 Ch 3 Sec 3-3
8	Output Mode: Overwrite vs. Append; Append for Snapshot Pipelines	1.25	LAB	TM-30 Ch 3 Sec 3-4
9	Scheduled Pipeline: Schedule Expression, Build Failure Email Alert	0.5	LAB	TM-30 Ch 3 Sec 3-5

Day 3 — Ontology Design

Block	Title	Hours	Method	Reference
10	Ontology Design Methodology: Domain Analysis, Entity ID, Relationship Mapping, Action Design	1.0	LEC	TM-30 Ch 4 Sec 4-1
11	Individual Design Exercise: Mission Requirement → Documented Ontology Schema	1.75	LAB	TM-30 Ch 4 Sec 4-2
12	Design Critique: Peer Presentations, Class Review Against 6-Item Rubric	2.0	WKS	TM-30 Design Rubric
13	Build the Approved Design: Create Ontology, Connect Pipeline via Write Step	2.25	LAB	TM-30 Ch 4 Sec 4-3

Day 4 — Analytics Tools and AIP Logic

Block	Title	Hours	Method	Reference
14	Contour: Pivot Tables, Calculated Columns, Parameter Controls, Saved Views	2.0	LAB	TM-30 Ch 5
15	Quiver: Multi-Object Analysis, Linked Views, Cross-Filter Propagation, Drilling	1.25	LAB	TM-30 Ch 6
16	AIP Logic Configuration: Triggers, Inputs, Outputs; Human Review Queue Design	1.5	LAB	TM-30 Ch 7 Sec 7-1
17	Data Lineage: Reading Lineage Graphs, Identifying Sources and Consumers	1.25	LAB	TM-30 Ch 8
18	C2DAO Production Standards: Quality Gates for Production-Ready Data Products	1.0	DIS	Standards Ch 3

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Day 5 — Governance and Practical Exercise

Block	Title	Hours	Method	Reference
19	Full C2DAO Promotion Workflow: Branch → Change → Submit → Respond → Approval	1.0	LAB	TM-30 Ch 9
20	Full-Stack Review: Raw Source → Pipeline → Ontology → Workshop → Governance	1.0	REV	TM-30 All Chapters
21	Practical Exercise Scenario Brief and Design Planning Time	1.25	BRF	—
22	Practical Exercise (Evaluated)	4.0	EVAL	TM-30 Practical Exercise Guide
	TM-30 Total	**40.0**		

3-4. TM-40G: ORSA Specialist

Course length: 5 days (40 hours) | **Evaluation:** Practical exercise (6 tasks); evaluated commander brief; Go/No-Go

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Day	Block	Title	Hours	Method	Reference
1	1	ORSA Role on MSS; Analytical Product Standards; Foundry Data Model	1.0	BRF	TM-40G Ch 1-2
1	2	Code Workspace Setup: Package Install, GPU/CPU Allocation, Reproducibility	2.0	LAB	TM-40G Ch 2 Sec 2-3
1	3	Foundry Dataset Connectivity: Reading via Spark/Pandas, Schema Inspection	0.75	LAB	TM-40G Ch 2 Sec 2-4
1	4	Writing Outputs to Foundry: Transaction Pattern for Results to Curated Datasets	2.0	LAB	TM-40G Ch 2 Sec 2-5
1	5	Data Profiling: Null Distributions, Outlier Detection, Feature Distributions	1.75	LAB	TM-40G Ch 3 Sec 3-1
2	6	Regression: Linear Regression for Readiness Forecasting, Validation Statistics	2.0	LAB	TM-40G Ch 3 Sec 3-2
2	7	Classification Models: Logistic Regression, Decision Trees, Cross-Validation	1.25	LAB	TM-40G Ch 3 Sec 3-3
2	8	Model Validation Standards: Residual Analysis, Documenting Assumptions	2.0	LAB	TM-40G Ch 3 Sec 3-4
2	9	Practice Build: Regression → Foundry Output → Quiver Visualization	1.75	LAB	TM-40G Ch 3
3	10	Time Series: Stationarity, ACF/PACF, ARIMA Model Identification	2.0	LAB	TM-40G Ch 4 Sec 4-1
3	11	ARIMA/SARIMA Build: Readiness Trend with 90% Confidence Bounds	1.25	LAB	TM-40G Ch 4 Sec 4-2
3	12	Monte Carlo: COA Comparison, Distribution Selection, 1,000-Trial Simulation	2.0	LAB	TM-40G Ch 5
3	13	Sensitivity Analysis; Logistics Stockage Risk Modeling	1.75	LAB	TM-40G Ch 5 Sec 5-3
4	14	Linear Programming: Resource Allocation Formulation, scipy/lpSolve	2.0	LAB	TM-40G Ch 6
4	15	Scheduling Optimization: Maintenance vs. Operational Commitments	1.25	LAB	TM-40G Ch 6 Sec 6-3
4	16	Wargame/Exercise Data Architecture: Collection Templates, Analysis Pipeline	2.0	LAB	TM-40G Ch 7
4	17	Quiver/Contour for ORSA: Forecast Dashboard, COA Comparison, Uncertainty	1.75	LAB	TM-40G Ch 8

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Day	Block	Title	Hours	Method	Reference
		Bounds			
5	18	Communicating Uncertainty: Confidence Intervals, Briefing Posture, Translation	1.0	LEC	TM-40G Ch 9
5	19	Common ORSA Brief Failures: Point Estimates Without Bounds, Methods-Paper Language	1.0	DIS	TM-40G Ch 9
5	20	Practical Exercise Scenario Brief and ORSA Product Standards Review	1.5	BRF	—
5	21	Practical Exercise (Evaluated): Regression + Time Series + Commander Brief	4.0	EVAL	TM-40G Practical Exercise Guide
		TM-40G Total	**40.0**		

3-5. TM-40H: AI Engineer

Course length: 5 days (40 hours)

NOTE

Day 1 Block 1 (AI Safety Seminar) is mandatory — no exceptions, no rescheduling.

Evaluation: Practical exercise (7 tasks); AIP Authorization Checklist review; Go/No-Go

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Day	Block	Title	Hours	Method	Reference
1	1	AI Safety: Human-in-the-Loop, OPSEC, Prohibited Use Cases, Army CIO Policy	2.0	SEM	TM-40H Ch 6; Appendix B
1	2	AIP Platform Architecture: Logic, Agent Studio, Code Workspaces, LLM Endpoints	1.75	LEC	TM-40H Ch 2
1	3	AIP Logic: First Workflow — Prompt, Input, Output Binding, Test Run	2.0	LAB	TM-40H Ch 3 Sec 3-1
1	4	AIP Logic: Conditional Chains, Error Handling, Structured JSON Output	1.75	LAB	TM-40H Ch 3 Sec 3-2
2	5	AIP Logic: Multi-Step Chains, Looping, Parallel Branches, Action Integration	2.0	LAB	TM-40H Ch 3 Sec 3-3
2	6	Python Transforms for AIP: Extracting Ontology Data, Context Formatting for Military Terminology	1.25	LAB	TM-40H Ch 4 Sec 4-1
2	7	Context Management: Chunking Strategies, Context Window Limits, Large Dataset Handling	2.0	LAB	TM-40H Ch 4 Sec 4-2
2	8	Ontology Integration: Write AIP Outputs via Actions; Human Review Queue for Uncertain Outputs	1.75	LAB	TM-40H Ch 4 Sec 4-3
3	9	RAG Architecture: Semantic Search, Retrieval from Ontology Objects, Context Construction	2.0	LAB	TM-40H Ch 5 Sec 5-1
3	10	RAG Pipeline Build: Retrieval → Context → Prompt → JSON Output → Ontology Write with Review Gate	1.25	LAB	TM-40H Ch 5 Sec 5-2
3	11	RAG Quality: Retrieval Relevance, Grounding Failures, Hallucination Detection	2.0	LAB	TM-40H Ch 5 Sec 5-3
3	12	End-to-End Workflow Practice: AIP Logic + RAG + Human Review + Ontology Write	1.75	LAB	TM-40H Ch 3-5
4	13	Agent Studio: Architecture, Tool Definition, Tool-Use Authorization, Memory Scope	2.75	LAB	TM-40H Ch 7
4	14	Agent Studio: Testing Tool Use; Confirming Authorization Controls	1.0	LAB	TM-40H Ch 7 Sec 7-4
4	15	Testing AI Outputs: Evaluation Frameworks, Red-Teaming, Adversarial Prompt Testing	2.0	LAB	TM-40H Ch 6 Sec 6-5

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Day	Block	Title	Hours	Method	Reference
4	16	AI Output Validation Framework; AIP Authorization Checklist Completion	1.75	LAB	TM-40H Appendix A, C
5	17	Production Deployment: Scheduling, Monitoring, Failure Alerting, Rollback	1.0	LAB	TM-40H Ch 8
5	18	Practical Exercise Scenario Brief and Workflow Design Time	1.0	BRF	—
5	19	Authorization Checklist Guidance; Evaluation Criteria Review	1.5	BRF	TM-40H Appendix A
5	20	Practical Exercise (Evaluated): Author → Test → Authorize → Deploy AIP Workflow	4.0	EVAL	TM-40H Practical Exercise Guide
		TM-40H Total	**40.0**		

3-6. TM-40I: ML Engineer

Course length: 5 days (40 hours) | **Evaluation:** Practical exercise (7 tasks); model card review; Go/No-Go

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Day	Block	Title	Hours	Method	Reference
1	1	MLE Role on MSS; Model Governance Overview; Responsible AI for Operational Models	1.0	BRF	TM-40I Ch 1, 9
1	2	Code Workspace Setup: GPU Allocation, Package Management, Foundry Connectivity	2.0	LAB	TM-40I Ch 2
1	3	Foundry Write Pattern for ML: Transaction-Based Output Writes from Code Workspace	0.75	LAB	TM-40I Ch 2 Sec 2-4
1	4	Feature Engineering Principles: Null Handling, Encoding, Scaling, Leakage Detection	2.0	LEC	TM-40I Ch 3 Sec 3-1
1	5	Feature Engineering Practice: Apply Standards to Provided Dataset; Document Features	1.75	LAB	TM-40I Ch 3 Sec 3-2
2	6	Feature Pipeline Build: Raw → Feature Matrix (Null, Encoding, Scaling)	2.75	LAB	TM-40I Ch 3 Sec 3-3
2	7	Feature Pipeline: Leakage Audit — Verify No Feature Derived from Label	0.75	LAB	TM-40I Ch 3 Sec 3-4
2	8	Feature Pipeline: Output to Foundry Curated Dataset via Write Transaction	2.0	LAB	TM-40I Ch 3 Sec 3-5
2	9	Experiment Setup: Train/Test Split, Cross-Validation, Baseline Model	1.75	LAB	TM-40I Ch 4 Sec 4-1
3	10	Model Training: scikit-learn/PyTorch in Code Workspace; Cross-Validation; Hyperparameter Tuning	2.75	LAB	TM-40I Ch 4 Sec 4-2
3	11	Model Evaluation: Accuracy/Precision/Recall/ROC-AUC; Acceptance Thresholds; Calibration	1.0	LAB	TM-40I Ch 5
3	12	Model Comparison Exercise: Train Two Models, Select Winner, Document Rationale	2.0	LAB	TM-40I Ch 5 Sec 5-3
3	13	Experiment Tracking: Log Parameters/Metrics to Foundry Model Registry; Versioning	1.75	LAB	TM-40I Ch 6
4	14	Model Deployment: Serving Endpoint, Inference API, Latency/Throughput Verification	2.0	LAB	TM-40I Ch 7 Sec 7-1
4	15	Connecting Deployed Model to Ontology: Actions That Invoke Inference, Write Predictions	1.25	LAB	TM-40I Ch 7 Sec 7-2
4	16	Monitoring Pipeline: Data Drift Detection, Threshold Definition, Alert Routing	2.0	LAB	TM-40I Ch 7 Sec 7-3

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Day	Block	Title	Hours	Method	Reference
4	17	Operational Use Case Patterns: Readiness Prediction, Logistics Forecasting, Anomaly Detection	1.75	LAB	TM-40I Ch 8
5	18	Model Governance: Model Card Completion — Assumptions, Limitations, Responsible AI Declaration	1.0	LAB	TM-40I Ch 9
5	19	Deployment Approval and C2DAO Governance for Deployed Models	1.0	BRF	Standards Ch 4
5	20	Practical Exercise Scenario Brief; Planning Time	1.5	BRF	—
5	21	Practical Exercise (Evaluated): Feature Pipeline → Train → Evaluate → Deploy → Monitor → Governance	4.0	EVAL	TM-40I Practical Exercise Guide
		TM-40I Total	**40.0**		

3-7. TM-40J: Program Manager

Course length: 3 days (24 hours) | **Evaluation:** Practical exercise (7 tasks); PM Dashboard Standards Checklist review; Go/No-Go

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Day	Block	Title	Hours	Method	Reference
1	1	Program Data Architecture: Program/Milestone/Resource/Risk Object Types — Design on Paper First	1.5	LAB	TM-40J Ch 2
1	2	Ontology: PM Object Types, Link Types (Program↔Milestone, Program↔Risk), Action Configuration	1.75	LAB	TM-40J Ch 2 Sec 2-4
1	3	Pipeline Builder: IMS Ingestion, DATEDIFF, Milestone Variance, RAG Status, Data-As-Of Timestamp	2.0	LAB	TM-40J Ch 3
1	4	Pipeline Builder: GFEBS Obligation Data in Append Mode; Run Twice, Verify Two Snapshot Records	2.0	LAB	TM-40J Ch 4 Sec 4-1
2	5	Workshop: Milestone Dashboard — RAG Conditional Formatting, Data-As-Of Timestamp Widget	2.0	LAB	TM-40J Ch 3 Sec 3-3
2	6	Quiver: Obligation Rate Chart, Reference Line at Quarterly Target, At-Risk Program Identification	1.25	LAB	TM-40J Ch 4 Sec 4-2
2	7	Contour: Portfolio Health Matrix — Cross-Program Roll-Up, Sort by overall_status Ascending	2.0	LAB	TM-40J Ch 5
2	8	Reporting: Scheduled Pipeline, Build Failure Notification, PDF Snapshot Export	1.25	LAB	TM-40J Ch 6
2	9	PM Dashboard Standards Checklist Walk-Through: Every Item, Common Failures	0.5	DIS	TM-40J Appendix A
3	10	Supervised Practice Run (Ungraded): Full Stack from Different Provided Dataset	3.25	LAB	TM-40J All Chapters
3	11	Practical Exercise (Evaluated)	4.0	EVAL	TM-40J Practical Exercise Guide
		TM-40J Total	**24.0**		

3-8. TM-40K: Knowledge Manager

Course length: 3 days (24 hours) | **Evaluation:** Practical exercise (6 tasks); PCS package instructor review (Day 3); Go/No-Go

Day	Block	Title	Hours	Method	Reference
1	1	KM Role on MSS; Knowledge Architecture Methodology; Why KM Systems Fail	1.0	BRF	TM-40K Ch 1
1	2	Ontology: Knowledge Object Types — Document, Lesson, AAR, SOP, ExpertiseProfile	2.0	LAB	TM-40K Ch 2
1	3	Workshop: AAR Submission Form — Required-Field Validation, Submission Confirmation	0.75	LAB	TM-40K Ch 3 Sec 3-2
1	4	Lessons Learned Pipeline: Intake, Deduplication, Tagging Taxonomy, Distribution Routing	2.0	LAB	TM-40K Ch 4
1	5	AIP Logic: Document Summarization; Human Review Queue — All AIP Outputs Begin as Draft	1.75	LAB	TM-40K Ch 5 Sec 5-1
2	6	Workshop: Knowledge Browser — Search, Filter by Tag/Unit/Date, Drill-Down	2.0	LAB	TM-40K Ch 5 Sec 5-4
2	7	SOP/Doctrine Version Control: Lifecycle, Version Tagging, Review Notification Workflow	1.25	LAB	TM-40K Ch 7 Sec 7-6
2	8	Personnel Expertise Mapping: ExpertiseProfile, Skills Taxonomy, Privacy Act Authorities	1.0	LAB	TM-40K Ch 8 Sec 8-1
2	9	PCS Knowledge Transfer: Key Person Dependency Analysis, Transfer Package Design	1.25	LAB	TM-40K Ch 9
2	10	AIP Prompt Iteration Lab: Test 5 Documents, Score Extraction, Revise, Retest	1.5	LAB	TM-40K Sec 5-3
3	11	PCS Package Draft Review: Each Trainee Presents; Instructor Reviews; Trainees Revise	2.25	WKS	TM-40K Ch 9
3	12	Practical Exercise Scenario Brief	0.75	BRF	—
3	13	Practical Exercise (Evaluated)	4.0	EVAL	TM-40K Practical Exercise Guide
		TM-40K Total	**24.0**		

3-9. TM-40L: Software Engineer

Course length: 5 days (40 hours) | **Evaluation:** Practical exercise (6 tasks); validator test suite (8 test cases); deployment checklist review; Go/No-Go

Day	Block	Title	Hours	Method	Reference
1	1	SWE Role on MSS; 5-Layer Data Stack; OSDK Architecture vs. Standard REST	1.0	LEC	TM-40L Ch 1
1	2	OSDK Setup: Auth Architecture, Client Init, Token Handling, First Object Query	2.0	LAB	TM-40L Ch 2 Sec 2-1
1	3	OSDK: Filtering and Sorting — Query Predicates, Sort Expressions, Field Selection	0.75	LAB	TM-40L Ch 2 Sec 2-2
1	4	OSDK: Pagination — ResourceIterator, Iterating All Pages, Multi-Page Result Sets	2.0	LAB	TM-40L Ch 2 Sec 2-3
1	5	OSDK: Link Traversal — Querying Related Objects Across Link Types	1.75	LAB	TM-40L Ch 2 Sec 2-4
2	6	OSDK: Action Execution — Async Response, Task ID Polling for Completion	2.0	LAB	TM-40L Ch 3 Sec 3-1
2	7	OSDK: Error Handling and Retry — Action Failures, Timeout, Structured Error Response	1.25	LAB	TM-40L Ch 3 Sec 3-2
2	8	OSDK: Object Subscriptions — Real-Time Change Notifications via WebSocket	2.0	LAB	TM-40L Ch 3 Sec 3-3
2	9	OSDK: Bulk Operations — Batch Queries, Bulk Action Submissions; Avoid Per-Object Loops	1.75	LAB	TM-40L Ch 3 Sec 3-4
3	10	Platform SDK: Dataset Read Operations, Write Transactions, File Resources, Branch Management	2.0	LAB	TM-40L Ch 4 Sec 4-1
3	11	Platform SDK Exercise: Build Dataset Integration Using Read/Write Transaction Pattern	1.25	LAB	TM-40L Ch 4 Sec 4-2
3	12	TypeScript FOO: Repository Structure, Computed Property Implementation, Function Registration	2.0	LAB	TM-40L Ch 5 Sec 5-1
3	13	FOO: Bulk Query Patterns — Avoiding N+1 Queries; Test with 200+ Object Set	1.75	LAB	TM-40L Ch 5 Sec 5-2
4	14	TypeScript Action Validators: Multi-Condition, Cross-Field Logic, Error Message Standards	2.0	LAB	TM-40L Ch 6 Sec 6-1
4	15	Validator Testing: Test Suite — 4 Valid, 4 Invalid; Each Paired with Expected Error Message	1.25	LAB	TM-40L Ch 6 Sec 6-2

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Day	Block	Title	Hours	Method	Reference
4	16	Slate Applications: Structure, Foundry API Integration, Widget Binding, Initial Data Load	2.0	LAB	TM-40L Ch 7 Sec 7-1
4	17	Slate: State Management on Action Completion; Error State Display for Failed Actions	1.75	LAB	TM-40L Ch 7 Sec 7-2
5	18	CI/CD: Repository Discipline, PR Workflow, Automated Testing, C2DAO Deployment Checklist	1.0	LEC	TM-40L Ch 8
5	19	Security and Compliance: Token Handling, Input Sanitization, OPSEC for App Code	1.0	LEC	TM-40L Ch 9
5	20	Practical Exercise Scenario Brief; Planning Time	1.5	BRF	—
5	21	Practical Exercise (Evaluated): OSDK → Validator → Slate UI → Deployment Checklist	4.0	EVAL	TM-40L Practical Exercise Guide
		TM-40L Total	**40.0**		

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CHAPTER 4 — TRAINING RESOURCES

4-1. Instructor Requirements

Course	Minimum Instructor Qualification	T:I Ratio
TM-10	TM-20 certified; 90 days active MSS use	10:1
TM-20	TM-30 certified; 6+ months Foundry build experience; able to troubleshoot all TM-20 labs	8:1
TM-30	TM-40 (any track) or C2DAO SME designation; able to conduct design critiques	6:1
TM-40A (Intel WFF)	TM-40A certified; G2/S2 Intel functional background; TM-30 proficiency	8:1
TM-40B (Fires WFF)	TM-40B certified; Fires/FSCOORD functional background; TM-30 proficiency	8:1
TM-40C (M2 WFF)	TM-40C certified; G3/S3 movement and maneuver background; TM-30 proficiency	8:1
TM-40D (SUST WFF)	TM-40D certified; G4/S4 sustainment background; TM-30 proficiency	8:1
TM-40E (PROT WFF)	TM-40E certified; Protection functional background; TM-30 proficiency	8:1
TM-40F (MC WFF)	TM-40F certified; Mission Command/G6 background; TM-30 proficiency	8:1
TM-40G	FA49 or equivalent ORSA background; TM-40G certified or C2DAO SME designation	4:1
TM-40H	AIP Logic authoring experience; C2DAO AI SME designation; TM-40H certified	4:1
TM-40I	ML production experience; TM-40I certified; C2DAO MLE SME designation	4:1
TM-40J	Program management background; TM-30 certified; GFEBS/IMS proficiency	6:1
TM-40K	Knowledge management background; TM-30 certified; AIP Logic configuration proficiency	6:1
TM-40L	Software engineering background; OSDK/Platform SDK proficiency; TM-40L certified	4:1

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4-2. Training Environment Requirements

Course	Access Level Required	Provisioning Lead Time
TM-10	MSS Viewer (standard)	5 duty days
TM-20	MSS Builder	5 duty days
TM-30	MSS Editor + AIP Logic configuration	7 duty days
TM-40A–F (WFF)	MSS Builder	5 duty days
TM-40G	Code Workspace (CPU or GPU) + standard Editor	7–10 duty days
TM-40H	AIP Logic authoring + Agent Studio	7–10 duty days
TM-40I	GPU-enabled Code Workspace	10+ duty days
TM-40J	MSS Builder	5 duty days
TM-40K	MSS Builder + AIP Logic configuration	5–7 duty days
TM-40L	OSDK developer access + developer token	10+ duty days

4-3. Training Aids and Materials

- Instructor presentation slides (per lesson plan)
- Training datasets (synthetic, stored in MSS Training Environment)
- Practical exercise scenario packages (printed + digital; prepared by instructor prior to course)
- TM reference materials (each trainee receives digital copy before Day 1)
- PM Dashboard Standards Checklist, ORSA Product Standards Checklist, C2DAO Deployment Checklist (printed)

CHAPTER 5 — EVALUATION STANDARDS

5-1. Go/No-Go Standards

Go requires all three: 1. All critical tasks completed independently (no instructor assistance, no hints) 2. All hard No-Go items passed (see 5-2 below) 3. Minimum task threshold met (see course syllabus)

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5-2. Hard No-Go Items (Automatic Failure)

Course	Hard No-Go Item
TM-10	Incorrect classification marking or export procedure
TM-20	Viewer-role test account can trigger Action or modify data
TM-30	Fatally-flawed Ontology design not corrected before build; promotion submitted without description
TM-40H	Any AIP workflow writes to production Objects without human checkpoint
TM-40I	Model calibration not performed; governance document missing required sections
TM-40J	Dashboard has no data-as-of timestamp
TM-40K	AIP workflow auto-publishes without human review gate
TM-40L	Hardcoded credential in application code; validator test suite not fully passing

5-3. No-Go Remediation

A trainee who receives No-Go must: 1. Receive documented counseling within 1 duty day (DA Form 4856 or equivalent) 2. Conduct remediation on failed tasks — minimum 4 hours for TM-10/20/40J/40K; minimum 8 hours for TM-30/40G/40H/40I/40L 3. Be re-evaluated within 10 duty days 4. A second No-Go requires C2DAO approval before a third evaluation

All remediation events are documented on the Individual Training Record.

5-4. Course Completion Documentation

Upon successful completion (Go): 1. Instructor updates the Unit Training Status Matrix (Appendix A of MTP) 2. Individual Training Record annotated with: course, date, evaluator name, Go/No-Go 3. For TM-30 and above: trainee's commander receives a completion notification

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APPENDIX A — COURSE HOURS SUMMARY MATRIX

Course	Lecture/Brief	Lab	Discussion/Review	Workshop/Seminar	Evaluation	Total
TM-10	1.5	5.5	—	—	1.0	8.0
TM-20	—	31.25	2.75	—	4.0	38.0*
TM-30	1.0	26.0	1.0	4.75	4.0	36.75*
TM-40A–F (each)	1.0	15.0	1.0	—	3.0	20.0*
TM-40G	2.0	30.0	1.0	—	4.0	37.0*
TM-40H	3.75	28.25	—	2.0	4.0	38.0*
TM-40I	1.0	31.0	—	—	4.0	36.0*
TM-40J	0.5	17.75	0.5	—	4.0	22.75*
TM-40K	1.0	14.25	—	2.25	4.0	21.5*
TM-40L	2.0	30.0	—	—	4.0	36.0*

*Remainder of scheduled hours are review periods and scenario briefs not separately categorized above.

NOTE

TM-40A–F are 3-day/24-hour courses requiring TM-30 as a prerequisite. The hours summary above reflects approximate method distribution; exact distribution varies by WFF track and is specified in the applicable WFF Syllabus. TM-50G–L course hour breakdowns are specified in the applicable advanced specialist Syllabi and are not reproduced here.

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APPENDIX B — AMENDMENT RECORD

Amendment	Date	Description	Approved By
Initial Publication	March 2026	Initial POI for MSS Training Program	C2DAO

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